Density-Aware Branching Update - 2026-04-29

Project: Image-Constrained Generative Modeling of Retinal Vasculature

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Summary

This update continued the retinal vasculature modeling work by improving the density-aware branching model. The previous density-aware model mainly used the local vessel-density map to adjust effective branching depth. That helped control how far branches could grow, but it did not directly guide where branches should go or where terminal nodes should concentrate.

The new update makes density affect branch placement more directly.

What Changed

1. Density-Guided Branch Direction

The `ConstrainedTreeGenerator` now compares nearby candidate branch directions and prefers endpoints that fall in higher-density regions of the reference density map.

This means the density map no longer only changes depth. It can now nudge branch growth toward areas where the reference image suggests more vessel structure.

2. Density-Based Branch Survival

The constrained generator now applies a density-based survival probability after a candidate branch passes the usual geometric checks.

Branches ending in high-density regions are more likely to be kept. Branches ending in sparse regions are more likely to be pruned. This should help terminal nodes concentrate more directly according to the reference density map.

3. Baseline Model Preserved

The baseline `RetinalTreeGenerator` was not changed. The new behavior is isolated to `ConstrainedTreeGenerator`, so comparisons between baseline, image-constrained, and density-aware models remain meaningful.

Tests Added

A new test file was added:

`tests/test_density_aware_branching.py`

The tests verify that:

- density-guided angle selection prefers a higher-density endpoint
- branch survival probability is higher in dense regions than sparse regions

All current tests pass:

```
Ran 4 tests
OK
```

Current Status

The model implementation and tests are complete.

The result-generation script was started, but the run was interrupted before final verification. Some output files may have been partially regenerated:

- `figures/fig2_model_comparison.png`
- `figures/fig3_density_terminals.png`
- `figures/fig4_evaluation_summary.png`
- `results/evaluation_results.csv`
- `results/evaluation_summary.csv`

Before treating these figures and CSV files as final, the full pipeline should be rerun from start to finish:

```
& 'C:\Program Files\PostgreSQL\16\pgAdmin 4\python\python.exe' run_results.py
```

Interpretation

This update directly addresses the main weakness identified in the previous progress report: the density-aware branch rule was too indirect. The revised model now uses density in three ways:

1. local density adjusts effective branching depth
2. local density nudges branch direction
3. local density affects branch survival

Together, these changes should make the density-aware model more responsive to image-derived spatial structure.

Next Step

Rerun the full 15-image evaluation and compare the updated density-aware metrics against the previous summary. The most important quantities to inspect are:

- density correlation
- terminal coverage score
- terminal count
- total vessel length

If density correlation improves without excessive loss of coverage or vessel complexity, this update is a useful modeling improvement.